

MPG No. 8.8.127 $m'm'm$ (m'mm' setting) [Type III, orthorhombic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & M_y & 0 \end{bmatrix}$$

$$M_y = M_{py}^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ 0 & M_{yyy} & 0 \\ 0 & M_{zzy} & 0 \\ 0 & 0 & M_{zzz} \\ 0 & 0 & 0 \\ M_{xxy} & 0 & 0 \end{bmatrix}$$

$$M_{xxy} = -M_{fay}^{(1)} - M_{fby}^{(1)} + M_{py}^{(1)}$$

$$M_{yyy} = 2M_{fay}^{(1)} + 3M_{py}^{(1)}$$

$$M_{zzz} = -M_{fay}^{(1)} + M_{fby}^{(1)} + M_{py}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & M_{zzz} & 0 \\ 0 & 0 & -\frac{M_{zzz}}{2} \\ 0 & 0 & 0 \\ -\frac{M_{xxy}}{2} & 0 & 0 \end{bmatrix}$$

$$M_{xxy} = 2M_{py}^{(2)} - 2T_{dxz}^{(1)}$$

$$M_{zzz} = 2M_{py}^{(2)} + 2T_{dxz}^{(1)}$$

$$\frac{M_{zzz}}{2} = M_{py}^{(2)} + T_{dxz}^{(1)}$$

$$\frac{M_{xxy}}{2} = M_{py}^{(2)} - T_{dxz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & M_{yzz} \\ 0 & 0 & 0 \\ M_{xyx} & 0 & 0 \end{bmatrix}$$

$$M_{yzz} = -M_{py}^{(3)} + T_{dxz}^{(2)}$$

$$M_{xyx} = M_{py}^{(3)} + T_{dxz}^{(2)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).